SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly. **Questions:**

1. Identify appropriate **HTML5 semantic elements** missing in the structure.

2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyse how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I Chunduri Naga Sai Srikanth (192311159) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Chunduri Naga Sai Srikanth

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, wellexplained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answers

1. Online Symposium Portal

1. Identify missing HTML5 semantic elements.

- <header> – Defines the page header.
- <main> – Defines the main content.
- <section> – Groups related content.
- <footer> – Defines the footer.

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Symposium Registration</title>
</head>
<body>
    <header>
        <h1>Online Symposium</h1>
    </header>

    <main>
```

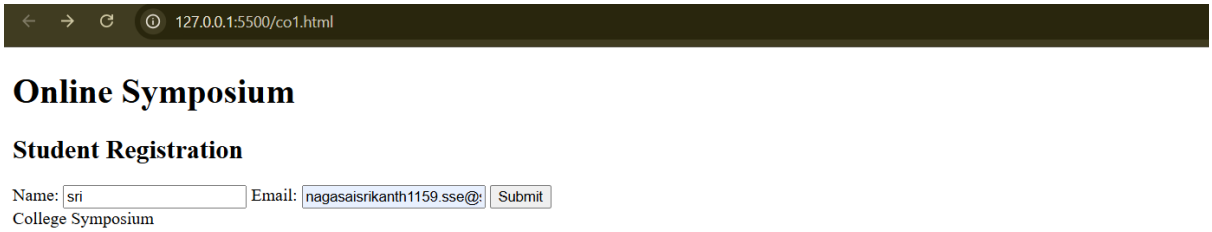
```
        <section>
            <h2>Student Registration</h2>
            <form action="/register" method="POST">
                <label for="name">Name:</label>
                <input type="text" id="name" name="name" required>

                <label for="email">Email:</label>
                <input type="email" id="email" name="email" required>

                <button type="submit">Submit</button>
            </form>
        </section>
    </main>

    <footer>College Symposium</footer>
</body>
</html>
```

Output:



3. Interpret the HTTP request.
   - The browser sends a POST request to /register.
   - Form data is sent in the request body.
   - The name and email fields are submitted.
   - The server processes the registration.
4. Analyze the HTTP response cycle.
   1. Browser sends the request.
   2. Server receives and validates the data.
   3. Server saves the registration if valid.
   4. Server sends an HTTP response, such as 200 OK or 201 Created.
   5. Browser displays the response or confirmation page.
5. Suggest usability and accessibility improvements.
   - Add labels for all input fields.
   - Use required for mandatory fields.
   - Use appropriate input types.
   - Provide clear error messages.
   - Ensure keyboard navigation and sufficient color contrast.

2: Cross-Browser Compatibility Issue

1. Identify improper nesting and missing closing tags.
   - The <h1> tag is not closed.
   - The <p> tag is not closed.
   - The document is missing <!DOCTYPE html> and the lang attribute.
2. Analyze how different browsers may interpret the structure.
   - Browsers may automatically insert missing closing tags.
   - The resulting DOM may differ from the intended structure.
   - Layout and styling can become inconsistent across browsers.
3. Identify missing semantic elements.
   - <header> – Page heading.
   - <main> – Main page content.
   - <section> – About Us content.
   - <footer> – Footer information, if needed.
4. Provide corrected W3C-compliant HTML.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Company</title>
</head>
```

```
<body>
    <header>
        <h1>Welcome</h1>
    </header>

    <main>
        <section>
            <h2>About Us</h2>
            <p>Welcome to our company website. </p>
        </section>
    </main>

    <footer>
        <p>Copyright 2026</p>
    </footer>
</body>
</html>
```

Output:



5. Suggest cross-browser compatibility techniques.

- Validate HTML using the W3C Validator.
- Use standard CSS and HTML.
- Test in Chrome, Firefox, Edge, and Safari.
- Use responsive design and CSS resets where appropriate.

3. Form Submission Failure

1. Identify issues in the form configuration.

- The button has type="button" instead of type="submit".
- The password is submitted using the GET method, which is insecure.
- Input fields lack labels and validation.

2. Analyze why the HTTP request is not triggered.

- A button with type="button" does not submit the form by default.
- Therefore, clicking Login does not automatically send the form data to the server.

3. Interpret the role of the GET method.

- GET sends form data in the URL as query parameters.
- URLs may be stored in browser history and server logs.
- It should not be used to transmit passwords.

4. Propose a corrected implementation.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
```

```
    <title>Login</title>
</head>
<body>
    <h1>Login Form</h1>

    <form action="/submit" method="POST">
        <label for="username">Username:</label>
        <input type="text" id="username"
               name="username" required>

        <label for="password">Password:</label>
        <input type="password" id="password"
               name="password" required>

        <button type="submit">Login</button>
    </form>
</body>
</html>
```

Output:

Login Form

Username: Password: Login

5. Suggest improvements for secure data transmission.

- Use HTTPS to encrypt data during transmission.
- Use the POST method for login credentials.
- Validate input on both the client and server.
- Store passwords using secure password hashing.
- Use secure session management and rate limiting.